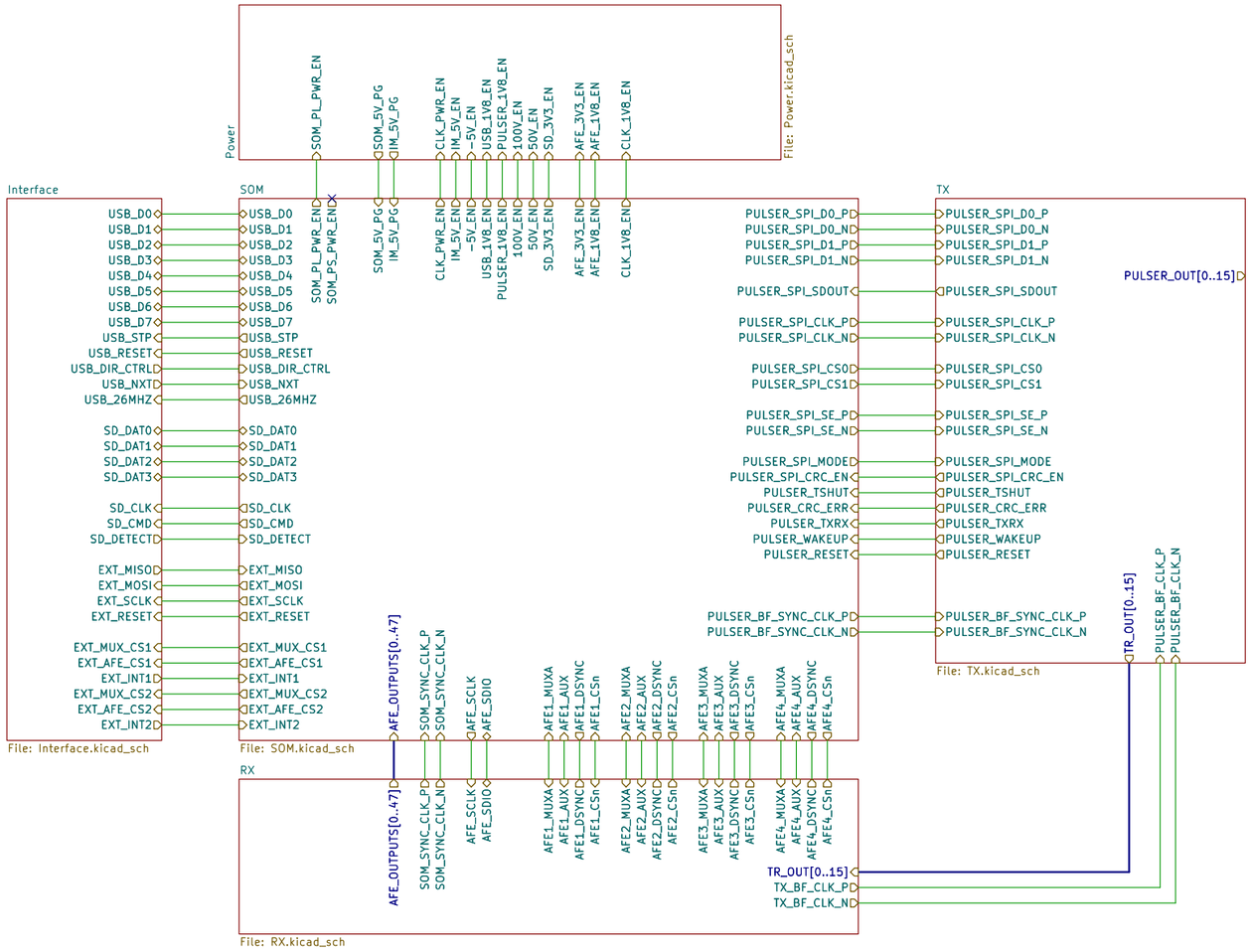


Power Adapter	12-24V (5A)	SOM Buck Converter	5V 4A (1.6A @ 12V)	Main SOM Supply (4A)	
		Intermediate Voltage Buck Converter	5V 2A (0.8A @ 12V)	Pulser Main Voltage (5V, 0.016A)	
				Pulser Negative Voltage (-5V, 0.023A)	
				AFE 1 & 2 Main Supply (3.3V, 0.26A)	
				AFE 1 & 2 Logic Supply (1.8V, 0.3A)	
				AFE 3 & 4 Main Supply (3.3V, 0.26A)	
				AFE 3 & 4 Logic Supply (1.8V, 0.3A)	
				Clock LDO Regulator 1	3.3V 0.1A
				Clock LDO Regulator 2	1.8V 0.1A
				External Interfaces LDO Regulator	3.3V ?A Up to 2.5A
		HDA Buck Converter	3.3V 1A (0.3A @ 12V)		
		HDB & HDC Buck Converter	3.3V 1A (0.3A @ 12V)		
		HPA Buck Converter	1.8V 1A (0.3A @ 12V)		
		USB & Pulser Buck Converter	1.8V 0.064A		
		SD Card Buck Converter	3.3V 1.2A (0.35A @ 12V)		
		+/-100V HV Converter	+/-100V 0.0005A (0.004A @ 12V)		
		+/-50V HV Converter	+/-50V 0.3A (1.2A @ 12V)		

U4A

PULSER_SPI_CLK_PD — H6 SCLKP
PULSER_SPI_CLK_ND — J6 SCLKM
PULSER_1V8
PULSER_SPI_SDOUT — H12 SDOUT
R1 10K
PULSER_SPI_DO_PD — H9 SDATAP_0
PULSER_SPI_DO_ND — J9 SDATAM_0
Only place resistors R1 & R2 if data line 1 is not used.
PULSER_SPI_D1_PD — H7 SDATAP_1
PULSER_SPI_D1_ND — J7 SDATAM_1
R2 10K
PULSER_SPI_SE_PD — H4 SENP
PULSER_SPI_SE_ND — J4 SENM
PULSER_SPI_MODE — G5 SPI_MODE
GND
PULSER_BF_SYNC_CLK_PD — L7 TR_BF_SYNCP
PULSER_BF_SYNC_CLK_ND — L6 TR_BF_SYNCM
PULSER_BF_CLK_PD — M7 BF_CLKP
PULSER_BF_CLK_ND — M6 BF_CLKM
PULSER_SPI_CS0 — F7 CHIP_SEL0
PULSER_SPI_CS1 — G7 CHIP_SEL1
PULSER_TXRX — H11 TX_RXZ
PULSER_SPI_CRC_EN — G4 EN_CRC
PULSER_TSHUT — H1 TSHUT
PULSER_WAKEUP — J12 WAKE_UP
PULSER_RESET — J1 RESET
PULSER_CRC_ERR — J2 CRC_ERR
R15 10K — GND
R16 10K — GND

OUT_1 A12 PULSER_OUT0
OUT_2 B12 PULSER_OUT1
OUT_3 C12 PULSER_OUT2
OUT_4 A11 PULSER_OUT3
OUT_5 B11 PULSER_OUT4
OUT_6 C11 PULSER_OUT5
OUT_7 A10 PULSER_OUT6
OUT_8 B10 PULSER_OUT7
OUT_9 A1 PULSER_OUT8
OUT_10 B1 PULSER_OUT9
OUT_11 C1 PULSER_OUT10
OUT_12 A2 PULSER_OUT11
OUT_13 B2 PULSER_OUT12
OUT_14 C2 PULSER_OUT13
OUT_15 A3 PULSER_OUT14
OUT_16 B3 PULSER_OUT15
PULSER_OUT[0..15]

HV Pulser Outputs

RX_1 L12 TR_OUT0
RX_2 M12 TR_OUT1
RX_3 L11 TR_OUT2
RX_4 M11 TR_OUT3
RX_5 L10 TR_OUT4
RX_6 M10 TR_OUT5
RX_7 L9 TR_OUT6
RX_8 M9 TR_OUT7
RX_9 L4 TR_OUT8
RX_10 M4 TR_OUT9
RX_11 L3 TR_OUT10
RX_12 M3 TR_OUT11
RX_13 L2 TR_OUT12
RX_14 M2 TR_OUT13
RX_15 L1 TR_OUT14
RX_16 M1 TR_OUT15
TR_OUT[0..15]

T/R Switch Outputs

NC F5
NC F6
NC G6
NC K1
NC K12

U4B

PULSER_5V — Logic 5V
E9 AVDDP_5
E8 AVDDP_5
E7 AVDDP_5
E6 AVDDP_5

PULSER_5VN — −5V
E4 AVDDM_5V
E5 AVDDM_5V

PULSER_1V8 — 1.8V
F8 AVDDP_1P8
F9 AVDDP_1P8
G8 AVDDP_1P8
G9 AVDDP_1P8

PULSER_100VN — −100V HV Supply
F1 AVDDM_HV_A
F2 AVDDM_HV_A
F12 AVDDM_HV_A
F11 AVDDM_HV_A
G1 AVDDM_HV_A
G2 AVDDM_HV_A
G11 AVDDM_HV_A
G12 AVDDM_HV_A

PULSER_50VN — −50V HV Supply
D1 AVDDM_HV_B
D2 AVDDM_HV_B
D11 AVDDM_HV_B
D12 AVDDM_HV_B
E1 AVDDM_HV_B
E2 AVDDM_HV_B
E11 AVDDM_HV_B
E12 AVDDM_HV_B

PULSER_100V — +100V HV Supply
A4 AVDDP_HV_A
A9 AVDDP_HV_A
B4 AVDDP_HV_A
B9 AVDDP_HV_A
C4 AVDDP_HV_A
C9 AVDDP_HV_A
D4 AVDDP_HV_A
D9 AVDDP_HV_A

PULSER_50V — +50V HV Supply
A5 AVDDP_HV_B
A8 AVDDP_HV_B
B5 AVDDP_HV_B
B8 AVDDP_HV_B
C5 AVDDP_HV_B
C8 AVDDP_HV_B
D5 AVDDP_HV_B
D8 AVDDP_HV_B

AVSS B6
AVSS B7
AVSS C3
AVSS C6
AVSS C7
AVSS C10
AVSS D3
AVSS D6
AVSS D7
AVSS D10
AVSS E3
AVSS E10
AVSS F3
AVSS F4
AVSS F10
AVSS G3
AVSS G10
AVSS H3
AVSS H5
AVSS H8
AVSS H10
AVSS J3
AVSS J5
AVSS J8
AVSS J10
AVSS K2
AVSS K3
AVSS K4
AVSS K5
AVSS K6
AVSS K7
AVSS K8
AVSS K9
AVSS K10
AVSS K11
AVSS L5
AVSS L8
AVSS M5
AVSS M8
GND

FLOATP_HV_A A6
FLOATM_HV_A J11
FLOATP_HV_B A7
FLOATM_HV_B H2

Output:
Indicates whether the pulser is in transmit or receiver mode.

PULSER_TXRX — R3 10K — C1 1nF — GND

PULSER_TSHUT — R4 10K — C2 1nF — GND
Output:
Thermal shutdown flag.

PULSER_CRC_ERR — R5 10K — C3 1nF — GND
Output:
SPI error flag (CRC error).

PULSER_100V
C37 100nF 200V
C38 100nF 200V
C39 100nF 200V
C40 100nF 200V
GND

PULSER_50V
C198 100nF 200V
C207 100nF 200V
C208 100nF 200V
C209 100nF 200V
GND

PULSER_100VN
C33 100nF 200V
C34 100nF 200V
C35 100nF 200V
C36 100nF 200V
GND

PULSER_50VN
C210 100nF 200V
C211 100nF 200V
C212 100nF 200V
C213 100nF 200V
GND

PULSER_5V
C23 100nF
C24 100nF
C25 100nF
C26 100nF
GND

PULSER_5VN
C21 100nF
C22 100nF
C20 100nF
C19 100nF
GND

PULSER_1V8
C41 100nF
C42 100nF
GND

Fanout Buffer 1

Feeds:
SOM sync signal
Pulser beamformer sync signal
AFE fanout buffer –>

Table 8-1. Input Selection Table

IN_SEL	ACTIVE CLOCK INPUT
0	IN0_P, IN0_N
1	IN1_P, IN1_N
Open	None (1)

72 MHz Crystal

Y1
ASFLMB-72.000MHZ-LY-T
C54 100nF
R11 50R
R104 1K
R105 1K

Bias for common mode voltage (0.9v)

U28
LMK1D1204RGTR

VDD, IN_SEL, VAC_REF0, IN0_P, IN0_N, IN1_P, IN1_N, EPAD, GND
OUT0_P, OUT0_N, OUT1_P, OUT1_N, OUT2_P, OUT2_N, OUT3_P, OUT3_N

TX_BF_CLK_P
TX_BF_CLK_N
SOM_SYNC_CLK_P
SOM_SYNC_CLK_N
R43 100R
R57 100R
R42 100R

It was recommended by the datasheet to terminate used & unused channels.

Fanout Buffer 2

Feeds:
4 * AFE

U29
LMK1D1204RGTR

Terminated at AFE side

CLKOUT_A0_P
CLKOUT_A0_N
CLKOUT_A1_P
CLKOUT_A1_N
CLKOUT_B0_P
CLKOUT_B0_N
CLKOUT_B1_P
CLKOUT_B1_N

C179 10uF
C175 100nF
C178 100nF
C215 10uF
C63 100nF
C235 10uF
C236 100nF
C237 100nF
C238 10uF
C70 100nF

C119 1uF
C120 1uF
C122 1uF
C123 1uF
C124 1uF
C126 1uF
C127 1uF
C128 1uF
C108 1uF
C110 1uF
C111 1uF
C112 1uF
C115 1uF
C116 1uF
C117 1uF
C118 1uF

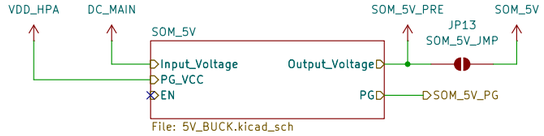
AFE1
File: RX_LANE.kicad_sch
AFE2
File: RX_LANE.kicad_sch
AFE3
File: RX_LANE.kicad_sch
AFE4
File: RX_LANE.kicad_sch

3V3D, 3V3A, 1V8D, 1V8A, CLK_P, CLK_N, AFE_DSYNC, AFE_SDIO, AFE_SCLK, AFE_CSn, AFE_MUXA, AFE_AUX
AFE_INA_P, AFE_INB_P, AFE_INC_P, AFE_IND_P
AFE_D0, AFE_D1, AFE_D2, AFE_D3, AFE_D4, AFE_D5, AFE_D6, AFE_D7, AFE_D8, AFE_D9, AFE_D10, AFE_D11

TR_OUT[0..15]
AFE_OUTPUTS[0..47]

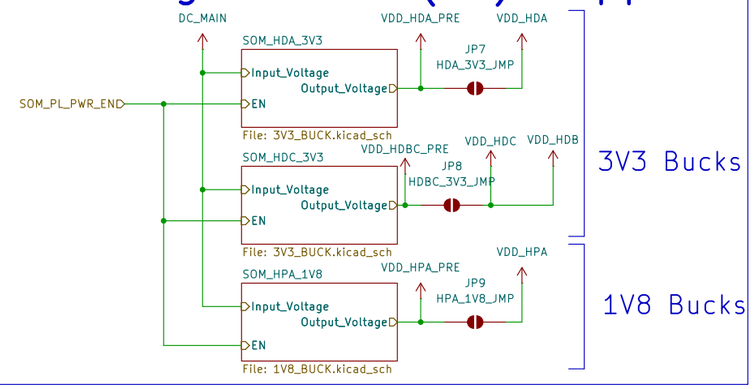
AFE IO are 3V3

Main SOM 5V Supply

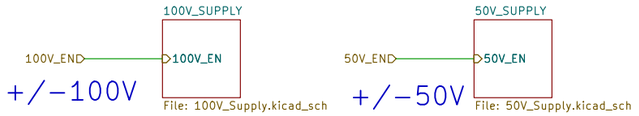


Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.20 – 1.50V	150 uA – 300 uA	External battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44

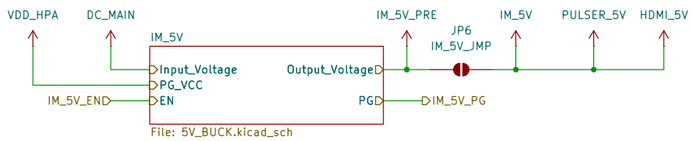
Processing Section (PL) Supplies



HV Supplies (+/- 50/100V)



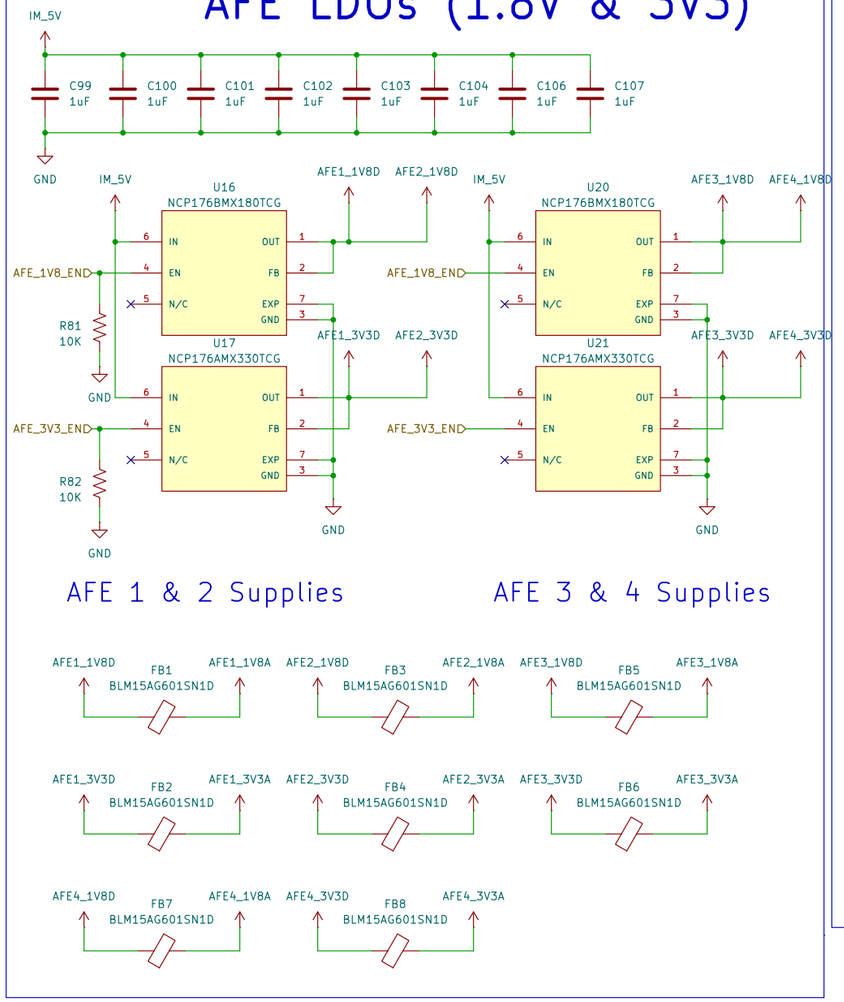
Pulser + Intermediate 5V Supply



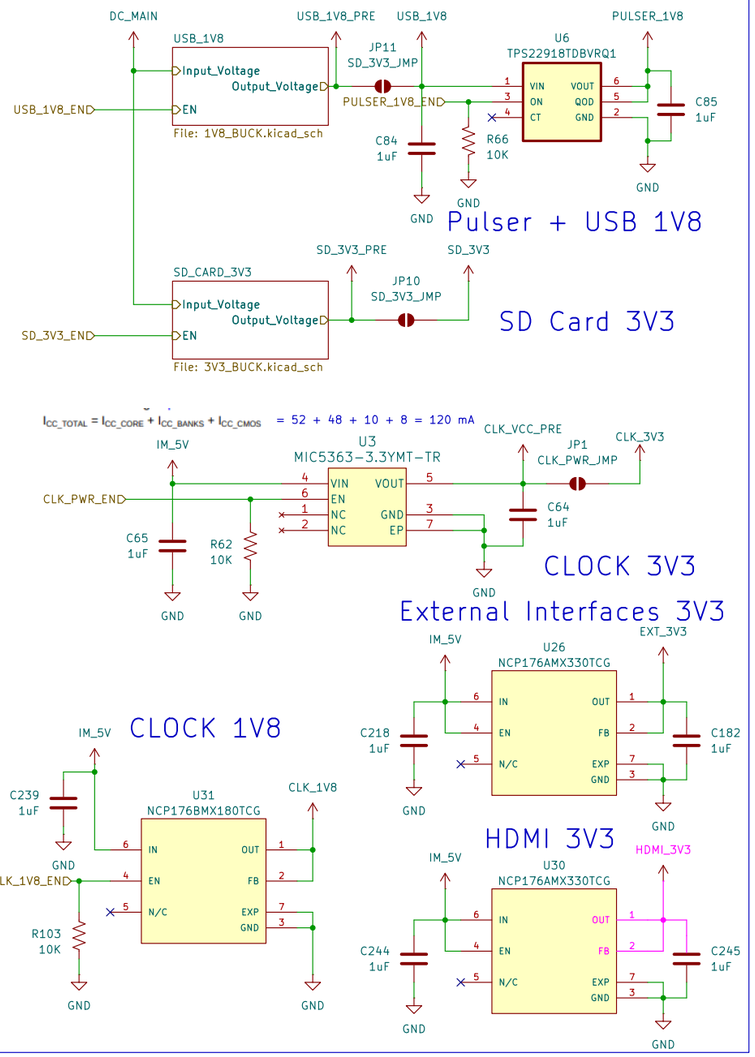
I_{WDD18}	Full channel mode	130	mA
I_{WDD33}	Full channel mode	130	mA
I_{VDD18}		22	mA
I_{VDD33}		2	mA

^ x4 = 1.2A

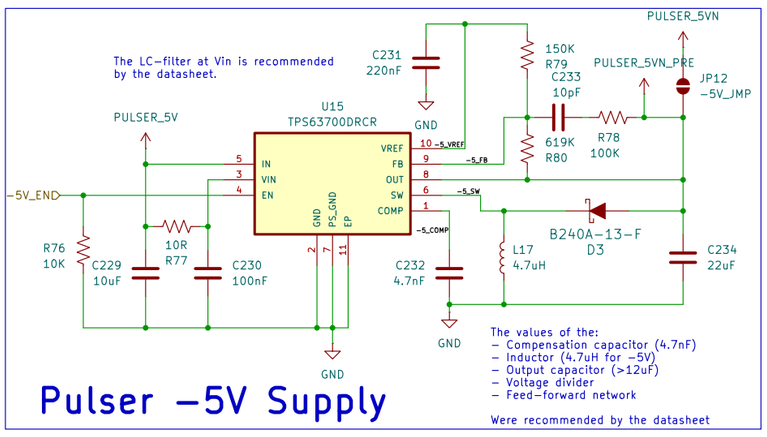
AFE LDOs (1.8V & 3V3)

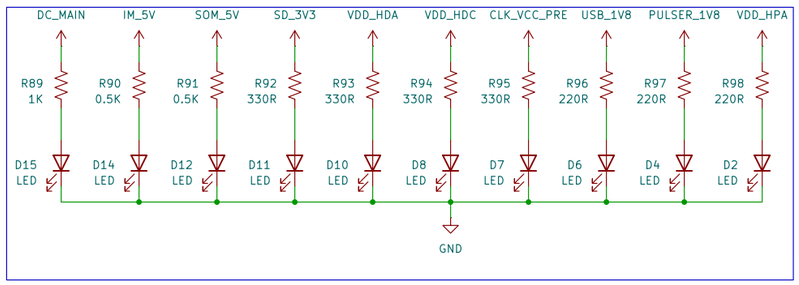


Miscellaneous Supplies



Pulser −5V Supply





VCC_BATT	PS BBRAM and real-time clock (RTC) supply voltage, requires external battery. Connect to GND when battery is not used.

J1A
REF-226081-01

J1B
REF-226081-01

J2A
REF-226081-01

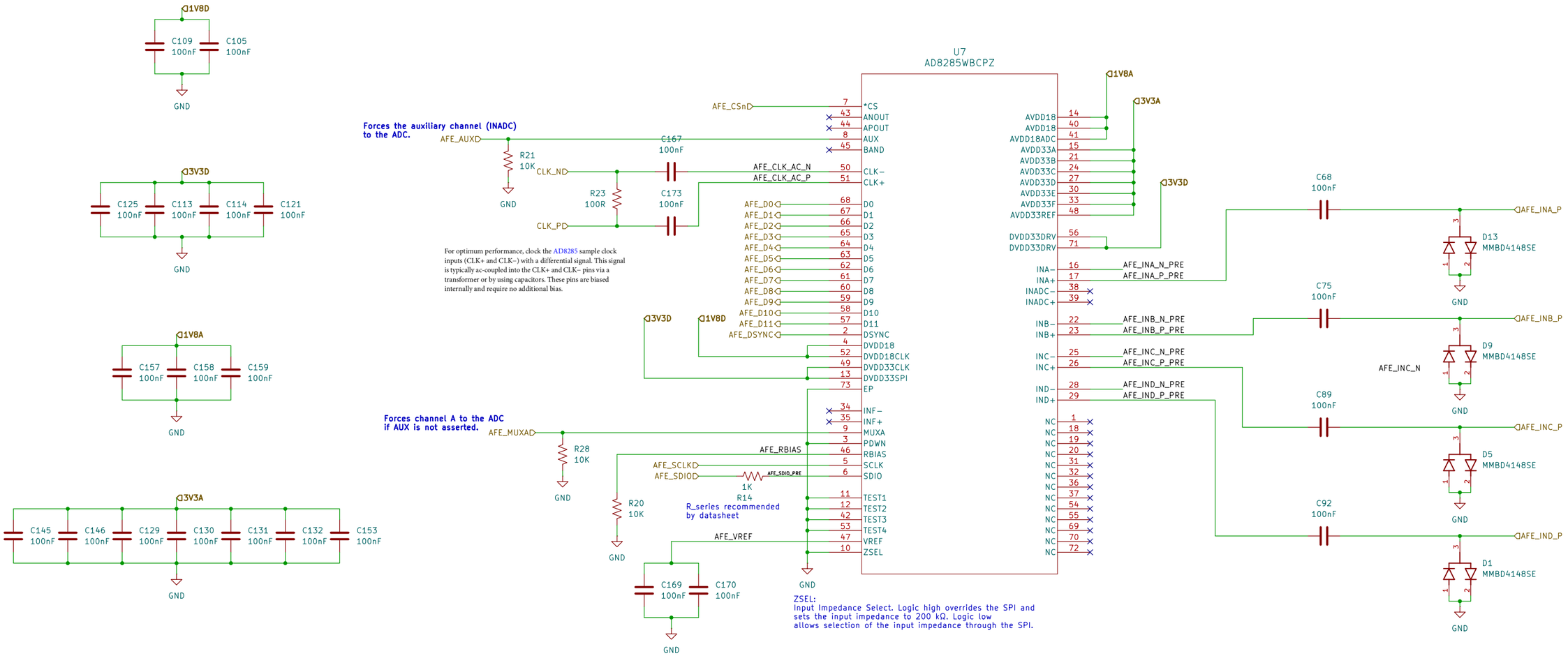
J2B
REF-226081-01

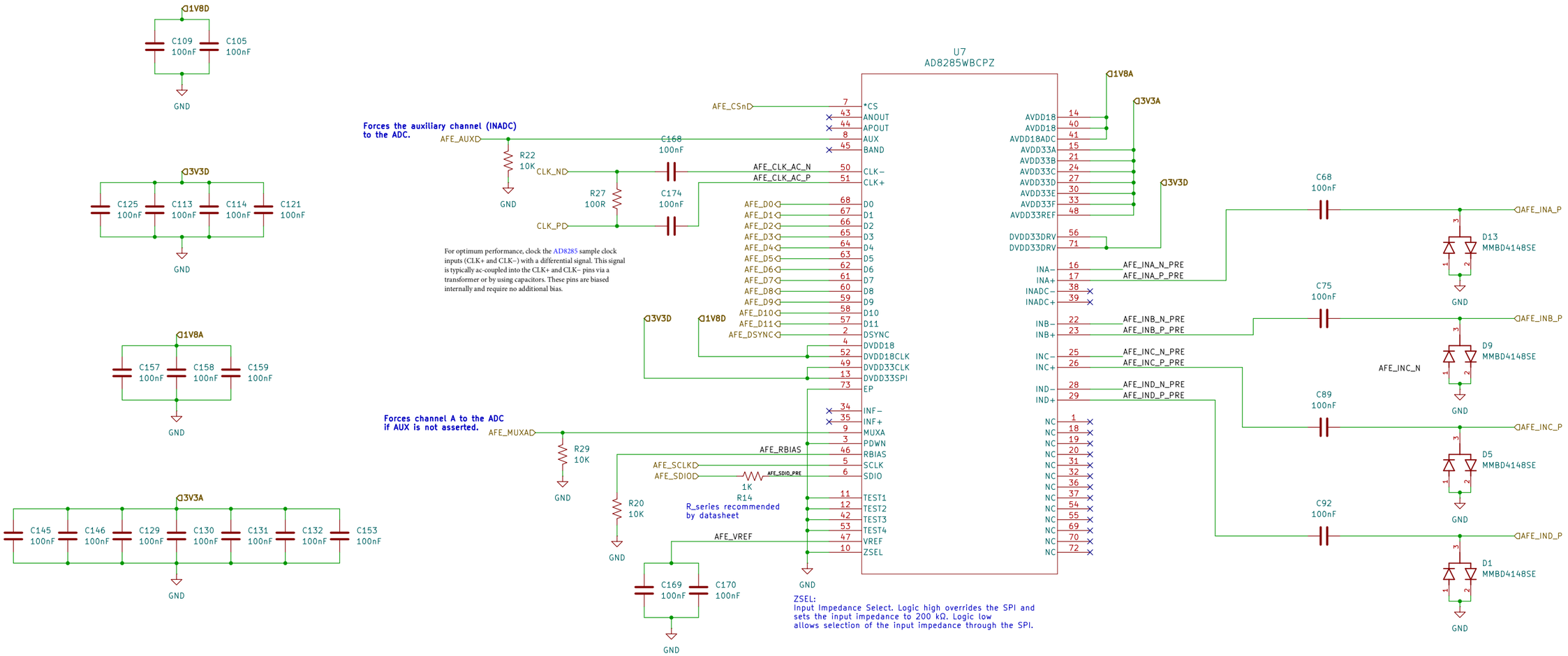
SOM240_1

SOM240_2

Boot Mode	PS_Mode Pins[3:0]	Physical Pin Location
Quad-SPI (32 bit)	0010	MIO[5:0]
eMMC	0110	MIO[22:13]

Reset Button


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R21 10K
GND
C167 100nF
CLK_N
R23 100R
C173 100nF
CLK_P
AFE_CLK_AC_N
AFE_CLK_AC_P
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_CSn
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R28 10K
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R_series recommended by datasheet
R20 10K
GND
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C169 100nF
C170 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
3V3D
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
NC
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_P
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R30 8R
R38 8R
R99 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R22 10K
GND
C168 100nF
CLK_N
R27 100R
C174 100nF
CLK_P
AFE_CLK_AC_N
AFE_CLK_AC_P
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_CSn
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R29 10K
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R_series recommended by datasheet
R20 10K
GND
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C169 100nF
C170 100nF
GND
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
3V3D
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
NC
C68 100nF
C75 100nF
C89 100nF
C92 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R34 8R
R40 8R
R100 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND

1V8D
C109 100nF
C105 100nF
GND

3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND

1V8A
C157 100nF
C158 100nF
C159 100nF
GND

3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND

U7
AD8285WBCPZ

Forces the auxiliary channel (INADC) to the ADC.

AFE_AUXD
R24 10K
GND
C171 100nF
CLK_ND
R31 100R
CLK_PD
C176 100nF

For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.

Forces channel A to the ADC if AUX is not asserted.

AFE_MUXAD
R32 10K
GND

AFE_SCLKD
AFE_SDIOD
AFE_SDIO_PRE
1K
R14
R_series recommended by datasheet
R20 10K
GND

AFE_VREF
C169 100nF
C170 100nF
GND

ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.

AFE_CSnD	7	*CS
	43	ANOUT
	44	APOUT
	8	AUX
	45	BAND
AFE_CLK_AC_N	50	CLK−
AFE_CLK_AC_P	51	CLK+
AFE_D0	68	D0
AFE_D1	67	D1
AFE_D2	66	D2
AFE_D3	65	D3
AFE_D4	64	D4
AFE_D5	63	D5
AFE_D6	62	D6
AFE_D7	61	D7
AFE_D8	60	D8
AFE_D9	59	D9
AFE_D10	57	D10
AFE_D11	2	D11
AFE_DSYNC	4	DSYNC
	52	DVDD18
	49	DVDD18CLK
	13	DVDD33CLK
	73	DVDD33SPI
	34	EP
	35	INF−
	9	INF+
	3	MUXA
AFE_RBIAS	46	PDWN
	5	RBIAS
	6	SCLK
	11	SDIO
	12	TEST1
	42	TEST2
	53	TEST3
	47	TEST4
	10	VREF
		ZSEL

	14	AVDD18
	40	AVDD18
	41	AVDD18ADC
	15	AVDD33A
	21	AVDD33B
	24	AVDD33C
	27	AVDD33D
	30	AVDD33E
	33	AVDD33F
	48	AVDD33REF
	56	DVDD33DRV
	71	DVDD33DRV
	16	INA−
	17	INA+
	38	INADC
	39	INADC+
	22	INB−
	23	INB+
	25	INC−
	26	INC+
	28	IND−
	29	IND+
	1	NC
	18	NC
	19	NC
	20	NC
	31	NC
	32	NC
	36	NC
	37	NC
	54	NC
	55	NC
	69	NC
	70	NC
	72	NC

1V8A
3V3A
3V3D

AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE

C68 100nF
D13 MMBD4148SE
AFE_INA_P
GND

C75 100nF
D9 MMBD4148SE
AFE_INB_P
GND

AFE_INC_N

C89 100nF
D5 MMBD4148SE
AFE_INC_P
GND

C92 100nF
D1 MMBD4148SE
AFE_IND_P
GND

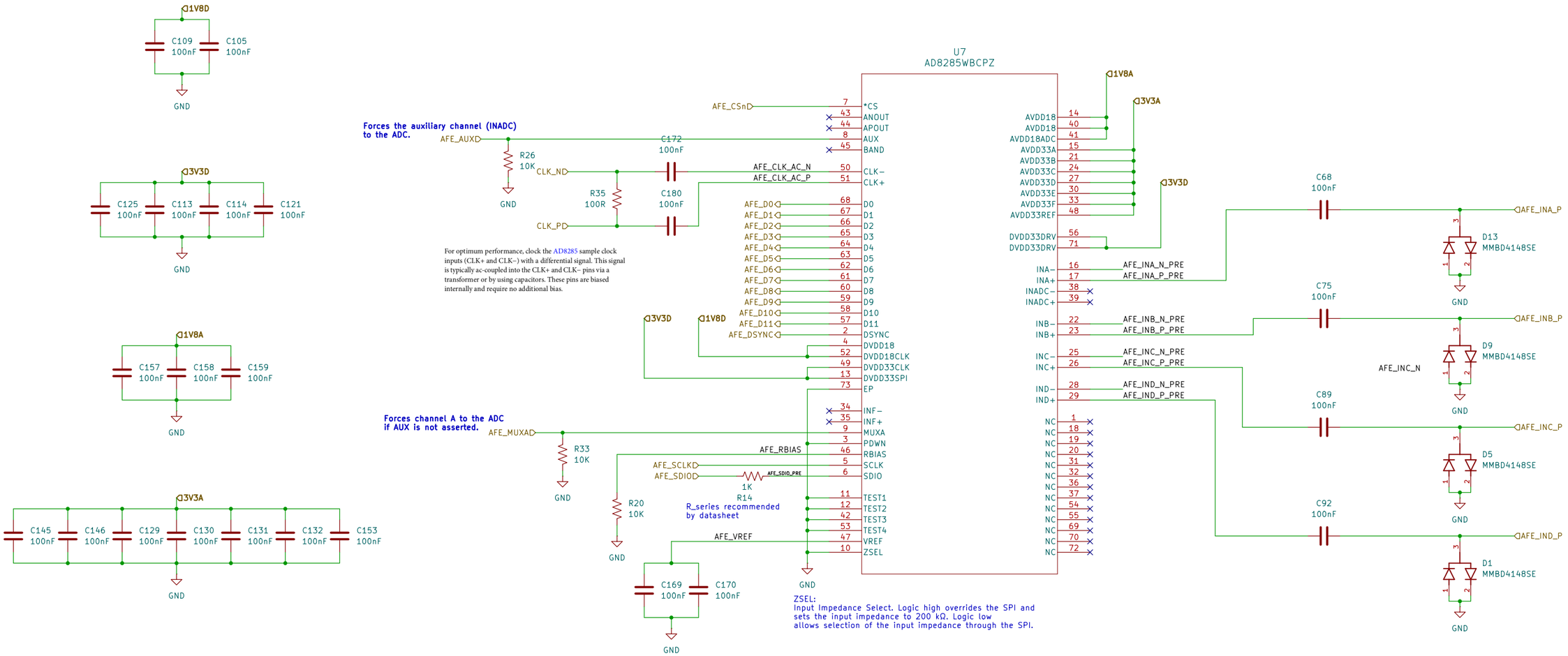
AFE_INA_N_PRE
R25 8R
C177 100nF
GND

AFE_INB_N_PRE
R36 8R
C181 100nF
GND

AFE_INC_N_PRE
R44 8R
C185 100nF
GND

AFE_IND_N_PRE
R101 8R
C189 100nF
GND

Change value according to pulser T/R switch out impedance


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R26 10K
GND
CLK_N
C172 100nF
R35 100R
C180 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R33 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R_series recommended by datasheet
R20 10K
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C169 100nF
C170 100nF
GND
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INC_N
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R37 8R
R45 8R
R102 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND

CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.

Check AN1310 for layout guidelines

3V3 & 1V5 are the outputs of internal LDOs

J7 TYPE-C-31-M-12 · U10 USBLC6-2SC6 · R61 10K · R109 50R · R110 5.1K · R111 5.1K · U13A TUSB1210BRHBR · U13B TUSB1210BRHBR · C47 47uF · C227 100nF · C228 100nF · C226 100nF · C214 100nF · C216 4.7uF · C217 4.7uF

USB HS PHY

U9 NXB0108BQ-Q100X · J6 External Board 1 · J5 External Board 2 · C240 1uF · C241 1uF · C242 1uF · C243 1uF

Common SPI & Control Lines
- MOSI
- MISO
- SCLK
- RESET

Seperate SPI Pins:
- AFE CS
- MUX CS

External Modules Connectors

U24 NVT4858HKZ · J3 47352-1001 · R65 100K · C165 1uF · C164 100nF · C166 100nF

Level conversion 3V3 <-> 1V8

SD Card

DP2HDMI

File: DP2HDMI.kicad_sch

C252 100nF · C253 100nF · C248 100nF · C249 100nF · C246 100nF · C247 100nF · C250 100nF · C251 100nF

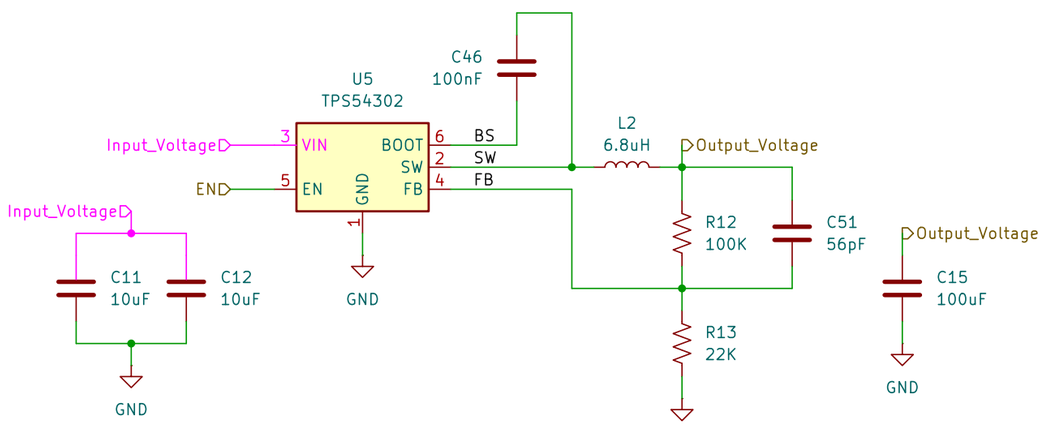
J4 2086581001

C254 100nF · C255 100nF · C256 100nF · C257 100nF · C258 100nF · C259 100nF · C260 100nF · C261 100nF

R69 3.3k · R70 3.3k · R85 10K · R67 10K · R87 10K · R107 10K · R88 10K · R68 10K · R86 10K · R106 10K · R17 4.02K · R83 3.3k · R84 3.3k

JP16 HPD_INV_PUP

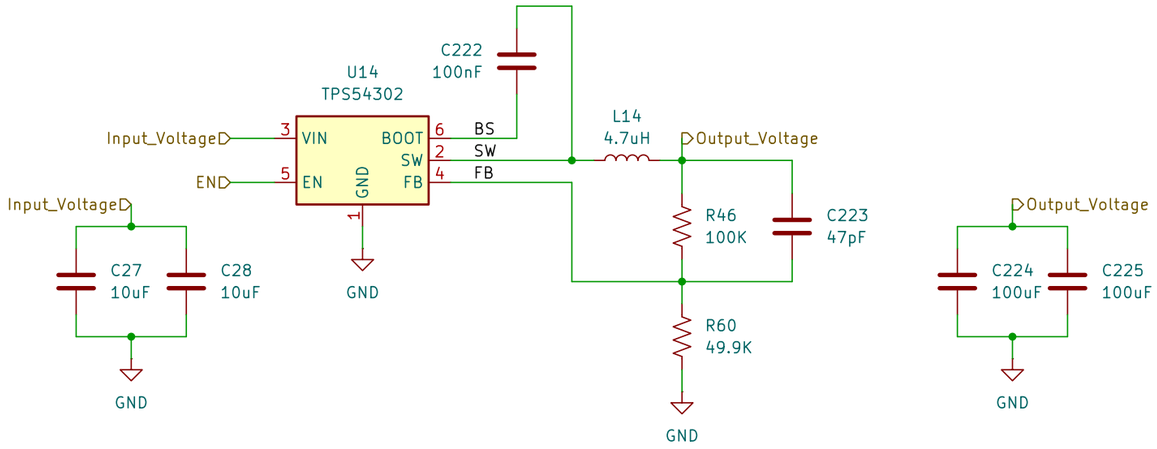
CR1 DT1240A-08LP3810-7



WEBENCH DESIGNER LINK (3.3V)

IC1
MCDP5200B0T

A1	GND_1	GND_37	G8
A2	UFP_L3N_SSRX2P	GND_38	G9
A3	UFP_L3P_SSRX2N	DVDDP9_4	G10
A4	GND_2	TEST	G11
A5	UFP_L2N_SSTX2P	DFP_CONFIG1	G12
A6	UFP_L2P_SSTX2N	GPIO7	G13
A7	RX_REXT	I2C_SCL_GPIO1	H1
A8	UFP_L1N_SSTX1N	I2C_SDA_GPIO0	H2
A9	UFP_L1P_SSTX1P	DVDD18_3	H3
A10	GND_3	DVDDP9_5	H4
A11	UFP_L0N_SSRX1N	GND_39	H5
A12	UFP_L0P_SSRX1P	GND_40	H6
A13	GND_4	GND_41	H7
B1	UFP_AUXP_SBU2	GND_42	H8
B2	GND_5	GND_43	H9
B3	GND_6	GND_44	H10
B4	GND_7	NC_3	H11
B5	GND_8	GPIO13	H12
B6	GND_9	GPIO12	H13
B7	RESETN	DFP_DDC_SDA	J1
B8	GND_10	GND_45	J2
B9	GND_11	DVDD18_4	J3
B10	GND_12	GND_46	J4
B11	GND_13	GND_47	J5
B12	GND_14	GND_48	J6
B13	SPI_HOLD	GND_49	J7
C1	UFP_AUXN_SBU1	GND_50	J8
C2	GND_15	GND_51	J9
C3	AVDDP9_UFP_1	DVDDP9_6	J10
C4	AVDDP9_UFP_2	NC_4	J11
C5	AVDDP9_UFP_3	GPIO15	J12
C6	XTAL_IN	GPIO14	J13
C7	XTAL_OUT	DFP_DDC_SCL	K1
C8	AVDDP9_UFP_4	CEC_GPIO10	K2
C9	AVDDP9_UFP_5	DFP_HPD_IN	K3
C10	AVDDP9_UFP_6	AVDD18_DFP_1	K4
C11	AVDDP9_UFP_7	AVDD18_DFP_2	K5
C12	SPI_CSN	GND_52	K6
C13	SPI_CLK	GND_53	K7
D1	GND_16	AVDD18_DFP_3	K8
D2	AVDDP9_UFP_8	AVDD18_DFP_4	K9
D3	AVDD18_UFP_1	DVDDP9_7	K10
D4	AVDD18_UFP_2	DVDD18_5	K11
D5	AVDD18_UFP_3	GPIO17	K12
D6	GND_17	GPIO16	K13
D7	REFCLK_OUT	GND_54	L1
D8	AVDD18_UFP_4	DFP_AUX_N	L2
D9	AVDD18_UFP_5	DFP_AUX_P	L3
D10	AVDD18_UFP_6	AVDDP9_DFP_1	L4
D11	DVDD18_1	AVDDP9_DFP_2	L5
D12	SPI_DI	AVDDP9_DFP_3	L6
D13	SPI_DO	AVDDP9_DFP_4	L7
E1	ALERTN_AHPD	AVDDP9_DFP_5	L8
E2	GND_18	AVDDP9_DFP_6	L9
E3	GND_19	DVDDP9_8	L10
E4	GND_20	DVDD18_6	L11
E5	GND_21	GPIO19	L12
E6	GND_22	GPIO18	L13
E7	GND_23	DFP_L3_HDMICLK_N	M1
E8	GND_24	GND_55	M2
E9	GND_25	GND_56	M3
E10	DVDDP9_1	GND_57	M4
E11	DVDD18_2	GND_58	M5
E12	SPI_WPN	GND_59	M6
E13	NC_1	GND_60	M7
F1	I2CM_SCL_PCONF1	GND_61	M8
F2	GND_26	GND_62	M9
F3	GND_27	GND_63	M10
F4	GND_28	GND_64	M11
F5	GND_29	GND_65	M12
F6	GND_30	GND_66	M13
F7	GND_31	DFP_L3_HDMICLK_P	N1
F8	GND_32	DFP_L2_HDMICH0_N	N2
F9	GND_33	DFP_L2_HDMICH0_P	N3
F10	DVDDP9_2	DFP_L1_HDMICH1_N	N4
F11	GPIO6	DFP_L1_HDMICH1_P	N5



WEBENCH DESIGNER LINK (1.8V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

n = 0.88

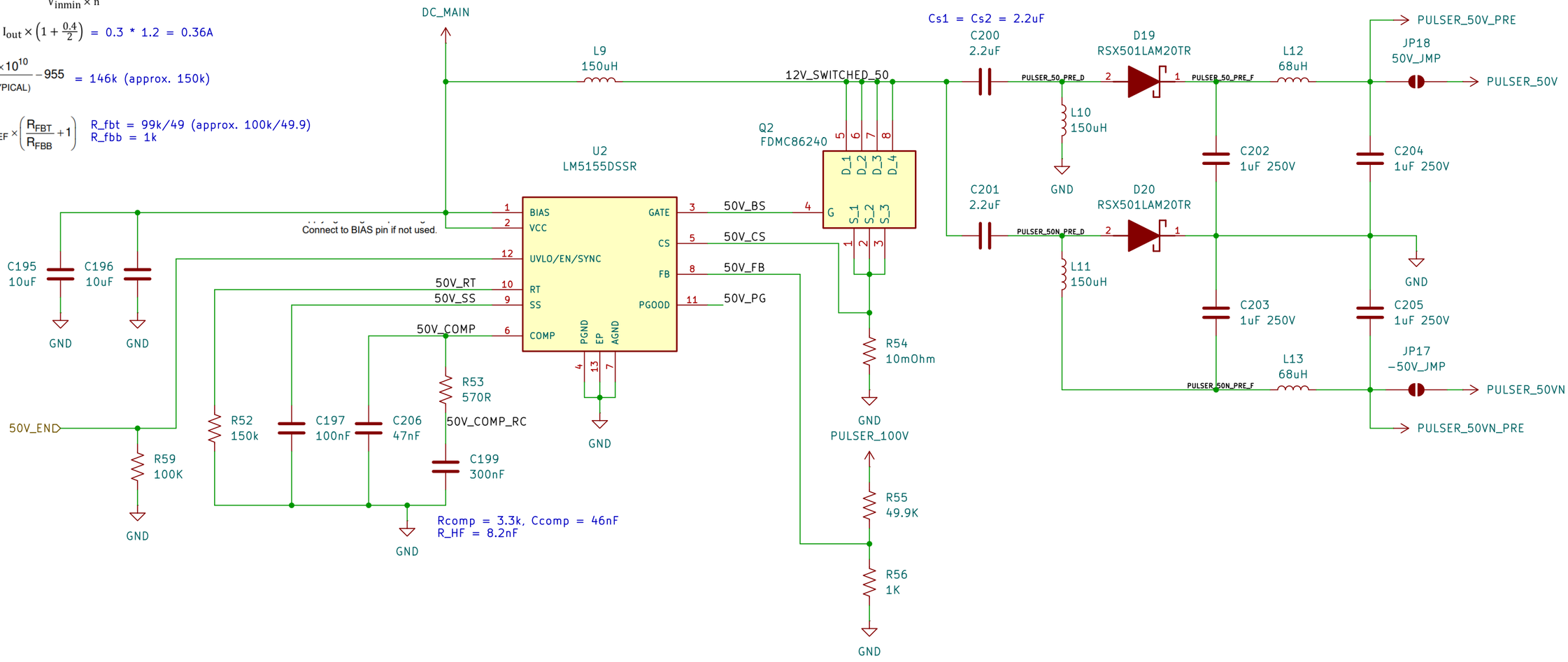
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

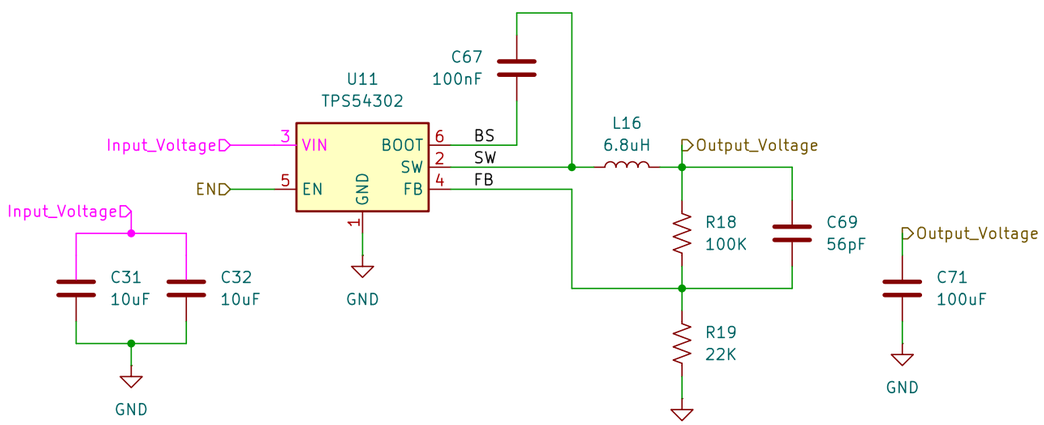
$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)

R_fbb = 1k





WEBENCH DESIGNER LINK (3.3V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

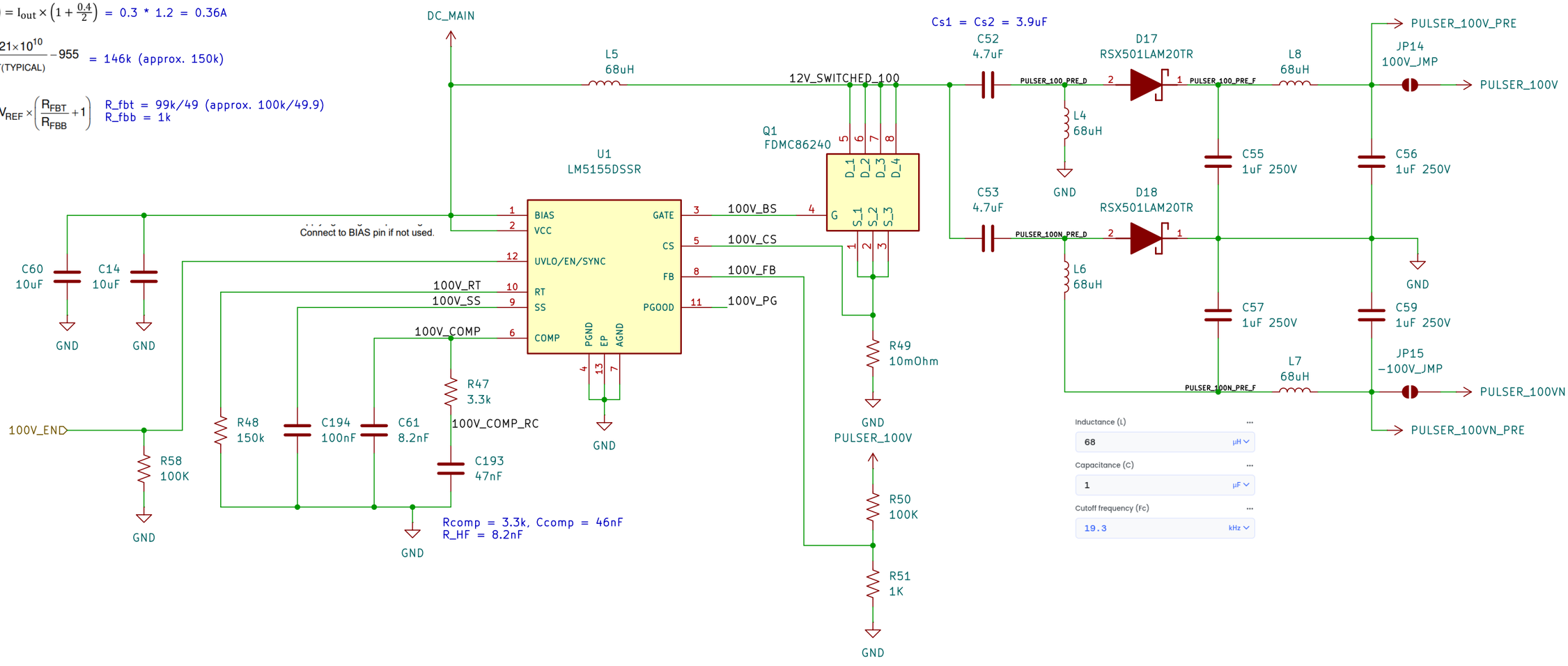
n = 0.88

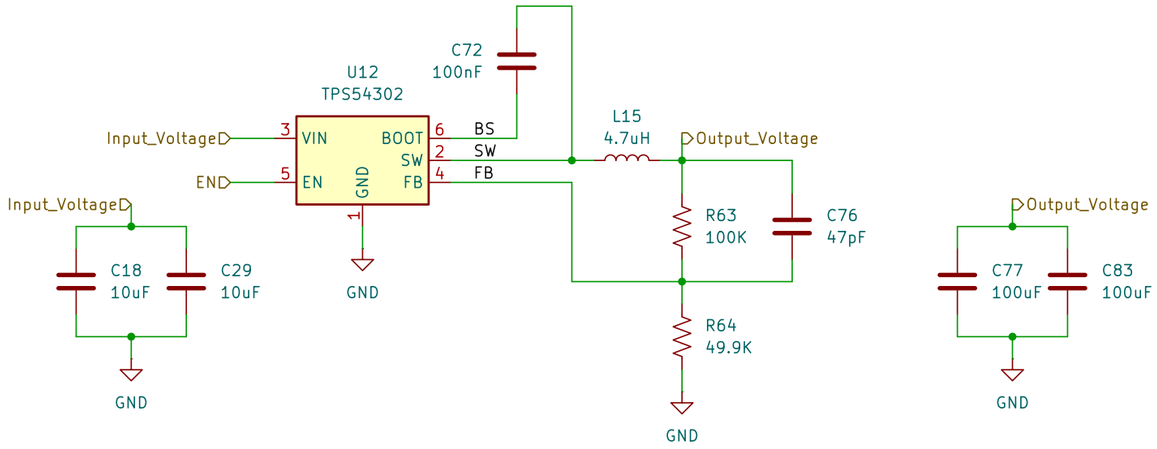
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

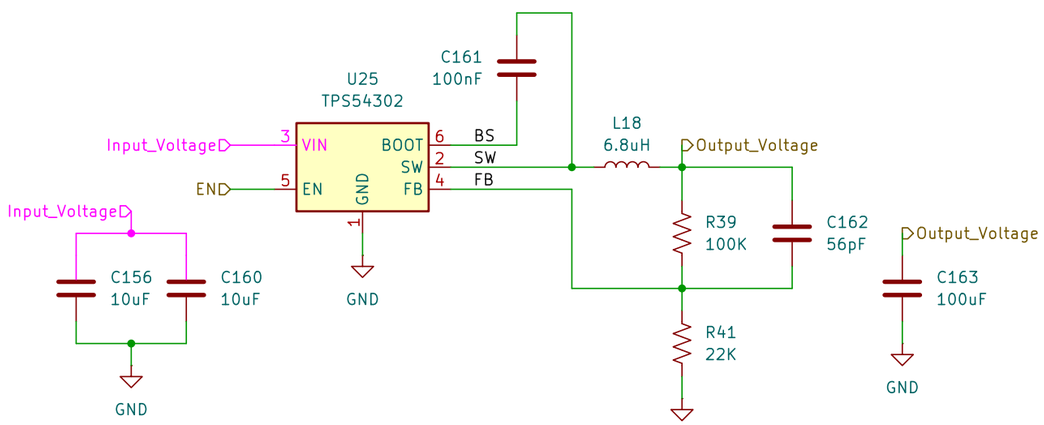
$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k

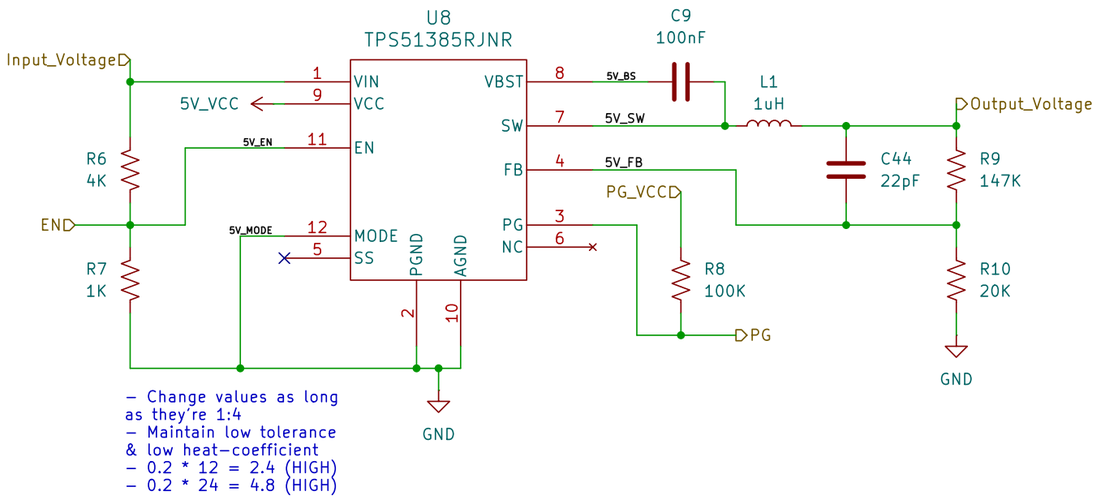


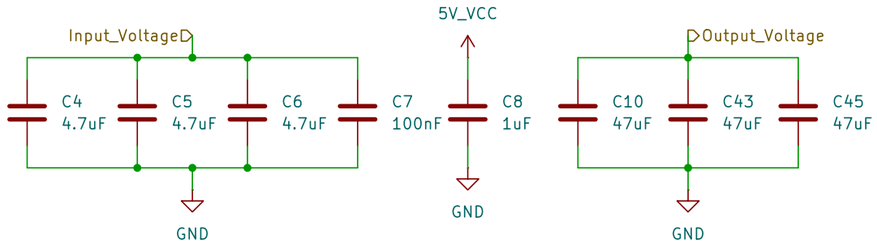


WEBENCH DESIGNER LINK (1.8V)

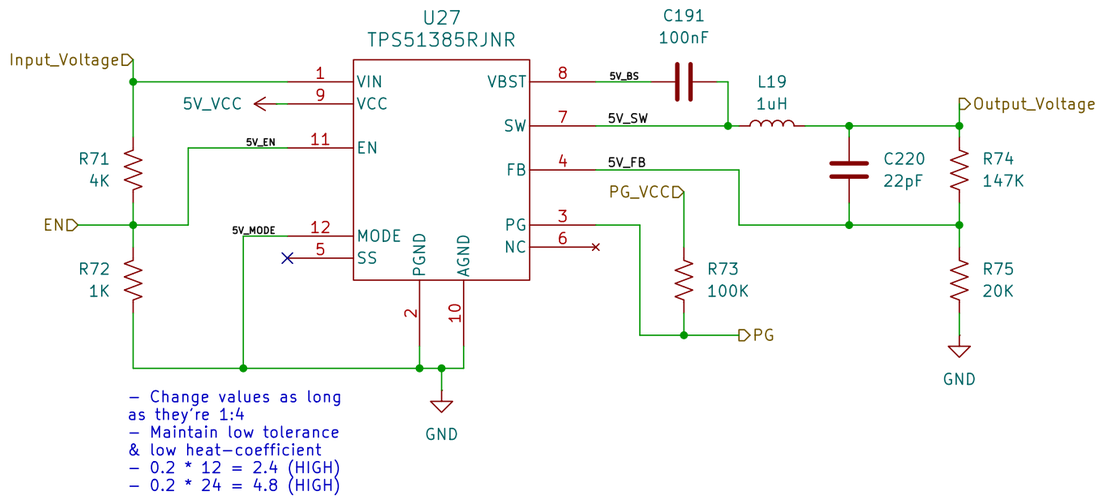


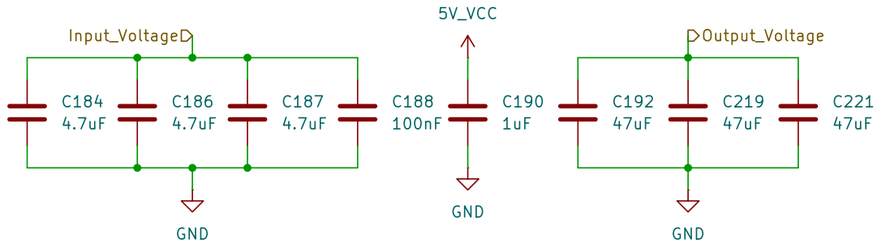
WEBENCH DESIGNER LINK (3.3V)





WeBench Designer Link





WeBench Designer Link